The Golden Parabola Heist

Whodunnit? · Inequalities · Teacher Guide and Answer Key

Unit	Time	Supplies	Group size
Inequalities	20–40 minutes	Pencil, scrap paper, scissors	Teams of 2–3 (or solo)

Skills practiced: Solving one-step, multi-step, compound and real-world inequalities.

General procedures

1. Give each student the **student sheet** (story and case board) and the **Show Your Work** page.
2. Print the **Clue Cards** (pages 3–7 of the student packet) and cut them apart on the dashed lines. Laminating them lets you reuse the set. Post them around the room.
3. Students visit the cards in any order. At each card they solve the problem on their work page, find their answer among the four choices, read the clue beside it, and cross things off the case board. **Every wrong choice leads to a false clue** that clears the real culprit, so a mistake shows up when the board does not narrow down to one suspect, one place and one method.
4. When a team thinks it knows, they bring you the three names. If they are wrong, let them hunt for the card where something went wrong.

Solution: Prof. Finch took it, in the Sculpture Garden, using this method: Tied to a balloon bouquet.

The reveal (read aloud): During the blackout, **Professor Finch** slipped into the **Sculpture Garden**, tied the golden statue under a **bouquet of balloons**, and floated it over the garden wall to a waiting van. Every alibi you cleared and every method you ruled out led you straight to him. Great detective work!

Clue map: what each correct clue clears

Card	Clears
1	Suspect: Chef Marco · Place: Café
2	Method: Packed in a catering cart
3	Suspect: Dr. Vega · Place: Basement Archive
4	Method: Hidden in a cello case
5	Suspect: Ms. Quill · Place: Grand Rotunda
6	Method: Rolled up in a rug
7	Suspect: Mr. Bishop · Place: Rooftop Terrace
8	Method: Swapped with a replica
9	Suspect: Ms. Tanaka · Place: Gift Shop
10	Method: Slipped into a delivery crate

Answer Key

The Golden Parabola Heist · Correct choice and worked solution for each card

#	Problem	Correct choice	Worked solution	Clue it leads to
1	$x + 7 < 3$	**B.** $x < -4$	$x + 7 < 3$ Subtract 7 from both sides: $x < 3 - 7$ $x < -4$	Chef Marco was plating desserts in the Café in front of a dozen guests.
2	$-4x \geq 20$	**D.** $x \leq -5$	$-4x \geq 20$ Divide both sides by -4 and flip the sign: $x \leq \frac{20}{-4}$ $x \leq -5$	The catering cart was searched at the door. No statue inside, so the Parabola was not packed in a catering cart.
3	$2x - 5 > 9$	**D.** $x > 7$	$2x - 5 > 9$ Add 5: $2x > 14$ Divide by 2: $x > 7$	Dr. Vega was studying old sketches in the Basement Archive. The sign-in sheet has her name all evening.
4	$\frac{x}{3} + 2 \leq 5$	**C.** $x \leq 9$	$\frac{x}{3} + 2 \leq 5$ Subtract 2: $\frac{x}{3} \leq 3$ Multiply by 3: $x \leq 9$	A guard checked every cello case at the exit. None held the Parabola.
5	$3(x - 2) \geq 2x - 1$	**A.** $x \geq 5$	$3(x - 2) \geq 2x - 1$ Distribute: $3x - 6 \geq 2x - 1$ Subtract $2x$: $x - 6 \geq -1$ Add 6: $x \geq 5$	Ms. Quill was giving the welcome speech in the Grand Rotunda. Every guest saw her.
6	$5 - 2x < 11$	**C.** $x > -3$	$5 - 2x < 11$ Subtract 5: $-2x < 6$ Divide by -2 and flip the sign: $x > -3$	The rug cleaners confirm nothing was rolled up in any gallery rug.
7	$-3 < 2x + 1 \leq 9$	**A.** $-2 < x \leq 4$	$-3 < 2x + 1 \leq 9$ Subtract 1 from all three parts: $-4 < 2x \leq 8$ Divide all three parts by 2: $-2 < x \leq 4$	Mr. Bishop was checking the cameras on the Rooftop Terrace. The video shows him the whole time.
8	$-8 \leq -2x < 6$	**C.** $-3 < x \leq 4$	$-8 \leq -2x < 6$ Divide all three parts by -2 and flip both signs: $4 \geq x > -3$ Write it smallest to largest: $-3 < x \leq 4$	The "replica" on display is just the old plaster model, so no swap took place.
9	$x - 4 < -3$ **or** $2x \geq 10$	**D.** $x < 1$ or $x \geq 5$	$x - 4 < -3$ or $2x \geq 10$ Piece 1: add 4: $x < 1$. Piece 2: divide by 2: $x \geq 5$ $x < 1$ or $x \geq 5$	Ms. Tanaka was buying postcards in the Gift Shop. The receipt is time-stamped 9:04 p.m.
10	$4 + 3x \leq 25$	**B.** $x \leq 7$	$4 + 3x \leq 25$ Subtract 4: $3x \leq 21$ Divide by 3: $x \leq 7$	Dock records show the delivery crate was empty, so the statue was never slipped into it.

Wrong-Answer Map

The Golden Parabola Heist · What a wrong choice looks like, and the false clue it leads to

Card	Wrong choice	Likely error	False clue
1	A. $x < 10$	added 7 instead of subtracting	Prof. Finch was signing books in the Café, and the line of fans stretched out the door.
1	C. $x > -4$	flipped sign without dividing by negative	Ms. Quill was leading a tour of the Sculpture Garden. Eight guests counted her footsteps.
1	D. $x < 4$	sign error moving 7	The balloon bouquet by the entrance was popped and checked. Nothing was tied to it, so the statue never floated out that way.
2	A. $x \le 5$	sign error in quotient	Chef Marco was setting out a snack tray in the Sculpture Garden. Guests grabbed mini quiches all through the blackout.
2	B. $x \ge 5$	sign error and wrong flip	Party planners say every gala balloon was tied on with ribbon only. No statue hung from a bouquet.
2	C. $x \ge -5$	forgot to flip when dividing by negative	Prof. Finch was giving a mini-lecture in the Grand Rotunda. The audience clapped at 9:02 p.m.
3	A. $x > 14$	forgot to divide by 2	Dr. Vega was sketching a bronze fox in the Sculpture Garden. Her sketchbook page is dated 9:05 p.m.
3	B. $x > 2$	subtracted 5 instead of adding	Prof. Finch was haggling over a poster in the Gift Shop. The cashier remembers the argument.
3	C. $x < 7$	flipped sign needlessly	The balloon bouquet was far too light and wobbly to carry anything heavy out the door.
4	A. $x \ge 9$	flipped sign needlessly	A guest photo of the balloon bouquet at 9:10 p.m. shows only ribbons and confetti.
4	B. $x \le 3$	divided instead of multiplying by 3	Mr. Bishop was walking his patrol route through the Sculpture Garden. The checkpoint scanner beeped at 9:01 p.m.
4	D. $x \le 21$	added 2 instead of subtracting	Prof. Finch was reading dusty ledgers in the Basement Archive. The sign-in sheet shows his signature.
5	B. $x \le 5$	flipped sign needlessly	The helium tank log shows all balloons were filled before the gala. No bouquet was added later.
5	C. $x \ge -7$	sign error combining constants	Ms. Tanaka was tuning her cello among the statues in the Sculpture Garden. Two guests stopped to listen.
5	D. $x \ge 1$	did not distribute to -2	Prof. Finch was pointing out constellations on the Rooftop Terrace to a group of guests.
6	A. $x > 3$	sign error in quotient	A photographer caught Ms. Quill posing with a donor in the Sculpture Garden right as the lights went out.
6	B. $x < 3$	sign error and no flip	Staff untied every balloon bouquet and found nothing hidden in the ribbons.
6	D. $x < -3$	forgot to flip when dividing by negative	A waiter swears Prof. Finch was stirring sugar into his coffee at the Café the whole time the lights were out.
7	B. $-1 < x \le 5$	added 1 instead of subtracting	Mr. Bishop locked the gate of the Sculpture Garden at 9:00 and stayed beside it. The gate log backs him up.
7	C. $-2 \le x < 4$	swapped the inequality symbols	The florist who made the balloon bouquets says they could never hold up a statue.
7	D. $-4 < x \le 8$	only subtracted 1, never divided	Prof. Finch was trying on a museum scarf in the Gift Shop. A photo from 9:03 p.m. proves it.
8	A. $-3 < x < 4$	lost the inclusive end	Security cameras show the balloon bouquet bobbing in the lobby all night. Nothing was tied to it.

8	B. $-3 \le x < 4$	mixed up which end is included	Chef Marco was lighting little lanterns along the paths of the Sculpture Garden when the power cut.
8	D. $-4 \le x < 3$	sign error dividing by -2	Prof. Finch was answering questions by the fountain in the Grand Rotunda with a crowd around him.
9	A. $x < -7$ or $x \ge 5$	subtracted 4 instead of adding	Prof. Finch was helping an intern sort boxes in the Basement Archive. The intern vouches for him.
9	B. $x > 1$ or $x \le 5$	flipped both signs	A kid popped the whole balloon bouquet at the party. No statue was attached.
9	C. $x < 1$ or $x \ge 20$	multiplied by 2 instead of dividing	Ms. Tanaka was playing a quiet tune in the Sculpture Garden. A dozen guests can hum it back.
10	A. $x < 7$	treated at most as strict inequality	Prof. Finch was taking selfies on the Rooftop Terrace. The photo timestamps all read between 9:00 and 9:10 p.m.
10	C. $x \le 21$	forgot to divide by 3	Dr. Vega was explaining a marble fountain to a student group in the Sculpture Garden.
10	D. $x \ge 7$	wrong direction	The Parabola weighs far more than any balloon bouquet could lift, so it did not leave that way.